CASE C — Ms. Sarah Collins

58-year-old Female · Non-CF | Bronchiectasis Exacerbation · Admitted with increased cough

Target: 70%+ | Physiotherapy

■ PROBLEM LIST

- Excess retained sputum in bronchiectatic airways
- Increased sputum viscosity and purulence
- Airflow obstruction secondary to mucus plugging
- Impaired mucociliary clearance
- Reduced respiratory muscle endurance
- Generalised deconditioning from infection/inflammation
- Exertional dyspnoea — reduced exercise tolerance
- Impaired ability to self-manage ACT effectively
- Fatigue limiting ADL performance
- Disrupted sleep from nocturnal cough
- Reduced participation in community activities

■ RED FLAGS

- ■ Haemoptysis (frank blood in sputum)
- Indicates possible erosion of bronchial artery — a known life-threatening complication of bronchiectasis. Volume matters: streaking vs. massive haemoptysis carry different urgency.
- Action: Stop ACT immediately. Do NOT perform percussion or vigorous expiratory techniques. Notify medical team immediately.
- ■ SpO■ <92% at rest / Acute severe dyspnoea
- Suggests acute respiratory failure — possible from mucus plugging, pneumothorax, or pneumonia superimposed on bronchiectasis. Physiotherapy cannot proceed safely.
- Action: Sit patient upright. Apply supplemental O■ as per prescription. Call medical emergency.
- ■ Systemic sepsis (fever >38.5°C, confusion, hypotension, tachycardia >120 bpm)
- Severe infective exacerbation requiring immediate medical management. The patient is not safe for active physiotherapy in this state.

■ PATHOPHYSIOLOGY

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■ DIFFERENTIAL DIAGNOSES

1. Community-Acquired Pneumonia (CAP) Supporting features Productive cough with purulent sputum over 2

- Productive cough with purulent sputum over 2 weeks
- Worsening dyspnoea and systemic fatigue
- CXR may show new consolidation

2. COPD Exacerbation with mucus hypersecretion Supporting features 58-year-old female with obstructive

- 58-year-old female with obstructive airway disease pattern
- Productive cough, dyspnoea, and reduced exercise tolerance
- Fatigue and reduced ACT efficacy

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Sputum Assessment (Colour, Consistency, Volume)

What it measures: Sputum quality and quantity — the primary physiotherapy treatment driver in bronchiectasis. Directs ACT technique selection, session frequency, and flags need for medical review. How to perform: Ask the patient to expectorate into a clear sputum pot after mornin

Peak Expiratory Flow Rate (PEFR) / Bedside Spirometry (FEV■)

What it measures: Degree of airflow obstruction — quantifies exacerbation severity, establishes baseline, and monitors physiotherapy response over the admission. How to perform: PEFR — patient seated upright, takes maximal inspiration, forms a tight seal around the mouthpiece, an

1-Minute Sit-to-Stand Test (1MSTS)

What it measures: Functional exercise capacity and physiological response to exertion — a validated, safe alternative to the 6MWT during acute exacerbation when a corridor walk may be too demanding. How to perform: Patient sits in a standard chair (approximately 45 cm height), ar

■ TREATMENT PLAN